Q1 - Single Correct

The circle with equation $x^2+y^2=1$ intersects the line $y=7x+5$ at two distinct points A and B. Let C the be point at which the positive X -axis intersects the circle. The $\angle ACB$, is

(1) $\tan^{-1}\left(\frac{4}{3}\right)$
(2) $\tan^{-1}\left(\frac{3}{4}\right)$
(3) $\tan^{-1}(1)$
(4) $\tan^{-1}\left(\frac{3}{2}\right)$

Q2 - Single Correct

In the XY -plane, the length of the shortest path from (0,0) to (12,16) that does not go inside the circle $(x-6)^2+(y-8)^2=25$, is

(1) $10\sqrt{3}$

(2) $10\sqrt{5}$

(3) $10\sqrt{3}+\frac{5\pi}{6}$

(4) $10\sqrt{3}+\frac{5\pi}{3}$

Q3 - Single Correct

ΔABC is right angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E, respectively. If $BD=20$ and $DC=16$, then the length AC equals (in units)

(1) $6\sqrt{21}$

(2) $6\sqrt{26}$

(3) 30

(4) 32

Q4 - Single Correct

A variable line moves in such a way that the product of the perpendiculars from $(a,0)$ and (0,0) is equal to k^2. The locus of the feet of the perpendicular from (0,0) upon the variable line is a circle, the square of whose radius is (given, $|a| < 2|k|$)

(1) $\frac{a^2}{4} + k^2$

(2) $\frac{a^2+k^2}{4}$

(3) $a^2 + \frac{k^2}{4}$

(4) $\frac{a^2+k^2}{2}$

Q5 - Single Correct

Let C be the circle of radius unity centred at the origin. If two positive numbers x_1 and x_2 are such that the line passing through $(x_1, -1)$ and $(x_2, 1)$ is tangent to C, then

(1) $x_1 x_2 = 1$

(2) $x_1 x_2 = -1$

(3) $x_1 + x_2 = 1$

(4) $x_1 x_2 = \frac{1}{4}$

Q6 - Single Correct

The locus of the mid-points of the chords of the circle $x^2 + y^2 + 4x - 6y - 12 = 0$ which subtend an angle of $\frac{\pi}{3}$ radians at its circumference is

(1) $(x-2)^2 + (y+3)^2 = 6.25$

(2) $(x+2)^2 + (y-3)^2 = 6.25$

(3) $(x+2)^2 + (y-3)^2 = 18.75$

(4) $(x+2)^2 + (y+3)^2 = 18.75$

Q7 - Single Correct

$\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right), \left(c, \frac{1}{c}\right)$ and $\left(d, \frac{1}{d}\right)$ are four distinct points on a circle of radius 4 units, then $abcd$ is equal to

(1) 4

(2) $\frac{1}{4}$

(3) 1

(4) 16

Q8 - Single Correct

A rhombus is inscribed in the region common to the two circles $x^2+y^2-4x-12=0$ and $x^2+y^2+4x-12=0$ with two of its vertices on the line joining the centres of the circles. The area of the rhombus is

(1) $8\sqrt{3}$ sq units

(2) $4\sqrt{3}$ sq units

(3) $16\sqrt{3}$ sq units

(4) None of these

Q9 - Single Correct

AB is diameter of a circle. CD is a chord parallel to AB and $2CD=AB$. The tangent at B meets the line AC produced at E, then AE is equal to

(1) AB

(2) $\sqrt{2}AB$

(3) $2\sqrt{2}AB$

(4) $2AB$

Q10 - Single Correct

$A(1,0)$ and $B(0,1)$ are two fixed points on the circle $x^2+y^2=1$. C is a variable point on this circle. As C moves, the locus of the orthocentre of the ΔABC, is

(1) $x^2+y^2-2x-2y+1=0$

(2) $x^2+y^2-x-y=0$

(3) $x^2+y^2=4$

(4) $x^2+y^2+2x-2y+1=0$

Q11 - Single Correct

The locus of the mid-points of the chords of the circle $x^2+y^2-2x-4y-11=0$ which subtend 60° at the centre is

(1) $x^2+y^2-4x-2y-7=0$

(2) $x^2+y^2+4x+2y-7=0$

(3) $x^2+y^2-2x-4y-7=0$

(4) $x^2+y^2+2x+4y+7=0$

Q12 - Single Correct

In a right ΔABC, right angled at A, on the leg AC as diameter, a semi-circle is described. The chord joining A with the point of intersection D of the hypotenuse and the semi-circle, then the length AC equals

(1) $\frac{AB \cdot AD}{\sqrt{AB^2+AD^2}}$

(2) $\frac{AB \cdot AD}{AB+AD}$

(3) $\sqrt{AB \cdot CD}$

(4) $\frac{AB \cdot AD}{\sqrt{AB^2-AD^2}}$

Q13 - Single Correct

B and C are fixed points having coordinates (3,0) and (-3,0), respectively. If the vertical $\angle BAC$ is 90°, then the locus of the centroid of ΔABC has the equation

(1) $x^2+y^2=1$

(2) $x^2+y^2=2$

(3) $x^2+y^2=\frac{1}{9}$

(4) $x^2+y^2=\frac{1}{9}$

Q14 - Multiple Correct

A family of linear function is given by $f(x)=1+c(x+3)$, where $c \in R$. If a member of this family meets a unit circle centred at origin at two coincident points, then c can be equal to

(1) $\frac{-3}{4}$

(2) 0

(3) $\frac{3}{4}$

(4) 1

Q15 - Multiple Correct

A circle passes through the points (-1,1),(0,6) and (5,5) . The points on this circle, the tangent(s) at which is/are parallel to the straight line joining the origin to its centre is/are

(1) (1,-5)

(2) (5,1)

(3) (-5,-1)

(4) (-1,5)

Q16 - Multiple Correct

Consider the circles $S_1 : x^2 + y^2 = 4$ and $S_2 : x^2 + y^2 - 2x - 4y + 4 = 0$. Which of the following statements are correct?

(1) Number of common tangents to these circles is 2

(2) If the power of a variable point P w.r.t. these two circles is same, then P moves on the line $x + 2y - 4 = 0$

(3) Sum of the y-intercepts of both the circles is 6

(4) The circles S_1 and S_2 are orthogonal

Q17 - Multiple Correct

If $\mathrm{al}^2 - \mathrm{bm}^2 + 2\mathrm{dl} + 1 = 0$, where a, b, d are fixed real numbers such that $\mathrm{a} + \mathrm{b} = \mathrm{d}^2$. Then the line $Ix + my + 1 = 0$ touches a fixed circle

(1) which cuts the X -axis orthogonally

(2) with radius equal to b

(3) on which the length of the tangent from the origin is $\sqrt{d^2-b}$

(4) None of the above

Q18 - Multiple Correct

Point M moved along the circle $(x-4)^2+(y-8)^2=20$. Then it broke away from it and moving along a tangent to the circle, cuts the X -axis at the point (-2,0) . The coordinates of the point on the circle at which the moving point broke away can be

(1) $\left(\frac{-3}{5}, \frac{46}{5}\right)$
(2) $\left(\frac{-2}{5}, \frac{44}{5}\right)$
(3) (6,4)
(4) (3,5)

Q19 - Multiple Correct

Three distinct lines are drawn in a plane. Suppose there exists exactly n circles in the plane tangent to all the three lines, then the possible values of n is/are

(1) 0
(2) 1
(3) 2
(4) 4

Q20 - Multiple Correct

Consider the circles,
$S_1: x^2+y^2+2x+4y+1=0$
$S_2: x^2+y^2-4x+3=0$
$S_3: x^2+y^2+6y+5=0$
Which of this following statements are correct?

(1) Radical centre of S_1, S_2 and S_3 lies in I quadrant
(2) Radical centre of S_1, S_2 and S_3 lies in IV quadrant

(3) Radius of the circle intersecting S_1, S_2 and S_3 orthogonally is 1

(4) Circle orthogonal to S_1, S_2 and S_3 has its x and y intercept equal to zero

Q21 - Multiple Correct

If $(a, 0)$ is a point on a diameter of the circle $x^2+y^2=4$. Then $x^2-4x-a^2=0$ has

(1) exactly one real root in (-1,0]

(2) exactly one real root in [2,5]

(3) distinct roots greater than 5

(4) distinct roots greater than - 1 and less than 5

Q22 - Multiple Correct

$C_1: x^2+y^2=25, C_2: x^2+y^2-2x-4y-7=0$ be two circles intersecting at the points A and B.

(1) Equation of common chord must be $x+2y-9=0$

(2) Equation of common chord must be $x+2y+7=0$

(3) Tangents at A and B to the circle C, intersect at $\left(\frac{25}{9}, \frac{50}{9}\right)$

(4) Tangents at A and B to the circle C_1 intersect at (1,2)

Q23 - Multiple Correct

Let L_1 be a line passing through the origin and L_2 be the line $x+y=1$. If the intercepts made by the circle $x^2+y^2-x+3y=0$ on L_1 and L_2 are equal, then the equation of L_1 can be

(1) $x+y=0$

(2) $x-y=0$

(3) $x+7y=0$

(4) $x-7y=0$

Q24 - Multiple Correct

Which of the following lines have the intercepts of equal length on the circle, $x^2+y^2-2x+4y=0$?

(1) $3x-y=0$

(2) $x+3y=0$

(3) $x+3y+10=0$

(4) $3x-y-10=0$

Q25 - Multiple Correct

Locus of intersection of the two straight lines passing through (1,0) and (-1,0) respectively and including an angle of 45° can be a circle with

(1) centre (1,0) and radius $\sqrt{2}$ units

(2) centre (1,0) and radius 2 units

(3) centre (0,1) and radius $\sqrt{2}$ units

(4) centre (0,-1) and radius $\sqrt{2}$ units

Q26 - Paragraph 1

Passage I (For Question 26, 27)

Let A, B and C be the sets of real numbers (x, y) defined as

$A:\{(x, y): y \geq 1\}$

$B:\left\{(x, y): x^{2}+y^{2}-4x-2y-4=0\right\}$

$C:\{(x, y): x+y=\sqrt{2}\}$

Number of elements in the $A \cap B \cap C$, is

(1) 0

(2) 1

(3) 2

(4) infinite

Q27 - Paragraph 1

$(x+1)^{2}+(y-1)^{2}+(x-5)^{2}+(y-1)^{2}$ has the value equal to

(1) 16

(2) 25

(3) 36

(4) 49

Q28 - Paragraph 2

Passage II (For Question 28, 29)

Consider 3 circles

$S_1 : x^2 + y^2 + 2x - 3 = 0$
$S_2 : x^2 + y^2 - 1 = 0$
$S_3 : x^2 + y^2 + 2y - 3 = 0$

The radius of the circle which bisects the circumferences of the circles $S_1 = 0$, $S_2 = 0$, $S_3 = 0$, is (in units)

(1) 2

(2) $2\sqrt{2}$

(3) 3

(4) $\sqrt{10}$

Q29 - Paragraph 2

If the circle $S = 0$ is orthogonal to $S_1 = 0, S_2 = 0$ and $S_3 = 0$ and has its centre at (a, b) and radius equals to r, then the value of $(a + b + r)$ equals

(1) 0

(2) 1

(3) 2

(4) 3

Q30 - Paragraph 3

Passage III (For Question 30, 31)

Consider the family of circles $x^2+y^2-2x-2\lambda y-8=0$, passing through two fixed points A and B. Also, $S=0$ is a circle of this family, the tangent to which at A and B intersect on the line $x+2y+5=0$

The distance between the points A and B is (in units)

(1) 4

(2) $4\sqrt{2}$

(3) 6

(4) 8

Q31 - Paragraph 3

The area of an equilateral triangle inscribed in S is (in sq units)

(1) $27\frac{\sqrt{3}}{4}$

(2) $9\frac{\sqrt{3}}{2}$

(3) $27\frac{\sqrt{3}}{2}$

(4) $9\sqrt{3}$

Q32 - Paragraph 4

Passage IV (For Question 32, 33, 34)

Consider a line pair $ax^2+3xy-2y^2-5x+5y+c=0$ representing perpendicular lines intersecting each other at C and forming a ΔABC with the X -axis.

If x_1 and x_2 are intercepts on the X -axis and y_1 and y_2 are the intercepts on the Y -axis, then the sum$(x_1+x_2+y_1+y_2)$ is equal to

(1) 6

(2) 5

(3) 4

(4) 3

Q33 - Paragraph 4

Distance between the orthocentre and circumcentre of ΔABC, is (in units)

(1) 2

(2) 3

(3) $\frac{7}{4}$

(4) $\frac{9}{4}$

Q34 - Paragraph 4

If the circle $x^2 + y^2 = 4y - k$ is orthogonal with the circumcircle of the ΔABC, then k equals

(1) $\frac{1}{2}$

(2) 1

(3) 2

(4) $\frac{3}{2}$

Q35 - Paragraph 5

Passage V (For Question 35, 36, 37)

Let C be a circle of radius r with centre O. Let P be a point outside C and D be a point on C. A line through P intersects C at Q and R, S is the mid-point of QR.

For different choices of line through P, the curve on which S lies, is

(1) a straight line

(2) an arc of the circle with P as centre

(3) an arc of the circle with PS as diameter

(4) an arc of circle with OP as diameter

Q36 - Paragraph 5

Let P is situated at a distance d from centre O, then which of the following does not equal to the product $(PQ) \cdot (PR)$?

(1) $d^2 - r^2$

(2) PT^2, where T is a point on C and PT is tangent to C

(3) $PS^2 - (QS)(RS)$

(4) $(PS)^2$

Q37 - Paragraph 5

Let XYZ be an equilateral triangle inscribed in C. If α, β, γ denote the distance of D from the vertices X, Y, Z respectively, the value of $(\beta + \gamma - \alpha)(\alpha + \gamma - \beta)(\alpha + \beta - \gamma)$, is

(1) 0

(2) $\frac{\alpha\beta r}{8}$

(3) $\frac{\alpha^3+\beta^3+\gamma^3-3\alpha b\gamma}{6}$

(4) None of these

Q38 - Integer Type

Number of value(s) of A for which the system of equations $x^2 = y^2$ and $(x - A)^2 + y^2 = 1$ has exactly 3 solutions, is

Q39 - Integer Type

If pair of tangents to a circle in first quadrant has equation $6x^2 - 5xy + y^2 = 0$ and if one point of contact is $(1, 2)$, where the radius of that circle is $\sqrt{\alpha} - \sqrt{\beta}$, then $\alpha - \beta$ equals

Q40 - Integer Type

Let C_1 and C_2 are circles defined by $x^2 + y^2 - 20x + 64 = 0$ and $x^2 + y^2 + 30x + 144 = 0$. The length of the shortest line segment PQ that is tangent to C_1 at P and C_2 at Q is k, then $\frac{k}{4}$ is

Q41 - Integer Type

Locus of middle points of a system of parallel chords with slope 2 of the circle $x^2+y^2-4x-2y-4=0$, has the equation $ax+by+c=0$, then $|a+b+c|$ is

Q42 - Integer Type

A variable circle C has the equation $x^2+y^2-2\left(t^2-3t+1\right)x-2\left(t^2+2t\right)y+t=0$, where t is a parameter. If the power of point $P(a, b)$ w.r.t. the circle C is constant, then the ordered pair $(a+b)$, is

Q43 - Integer Type

The centre of variable circle $x^2+y^2+2gx+2fy+c=0$ lies on the line $2x-2y+9=0$ and the variable circle cuts the circle $x^2+y^2=4$ orthogonally. If the variable circle passes through two fixed points (a, b) and (c, d), where $(b<d)$, then the value of $(2b+d)$ is equal to

Q44 - Integer Type

The graph of $x^2+y^2=4+12x+6y$ and $x^2+y^2=k+4x+12y$, where k satisfies $a \leq k \leq b$ for intersecting circles and for no other values of k. The value of $|b+3a|$ is

Q45 - Integer Type

Let C_1 and C_2 are circles defined by $x^2+y^2-20x+64=0$ and $x^2+y^2+30x+144=0$. The length of the shortest line segment PQ that is tangent of C_1 at P and C_2 at Q is l, then the value of $(l-15)$, is

Q46 - Integer Type

Let C be a circle that intersects each of the circles $(x+2)^2+y^2=4$, $(x-4)^2+(y-2)^2=4$ and $(x-4)^2+(y+2)^2=4$ in exactly one point does not contain any of these circles inside it. If the radius r of C is of the form p/q of where p and q are related prime, then $(p+q)$ is

Q47 - Integer Type

Two perpendicular tangents are drawn to the circle $x^2+y^2=1$, then the maximum value of square of ordinate of the point from where tangents are drawn is

Q48 - Integer Type

Six points $(x_i, y_i)\ i=1,2,3,\cdots;6$ are taken on circle $x^2+y^2=4$ such that $\sum_{i=1}^{6} x_i=8$ and $\sum_{i=1}^{6} y_i=4$. The line segment joining orthocentre of triangle made by any three points and the centroid of the triangle made by other three points passes through a fixed points (h, k) then $(h+k)$ is

Q49 - Integer Type

Let $f(x, y)=0$ be the equation of circle. If $f(0, \lambda)=0$ has equal roots $\lambda=1,1$ and $f(\lambda, 0)=0$ has roots $\lambda=\frac{1}{5}, 5$ then $(5r-4)$ is (where, r is radius of circle)

Q50 - Integer Type

If $\left(m_i, \frac{1}{m_i}\right), m_i>0, i=1,2,3,4$ are four distinct points on a circle, then the value of $\prod_{i=1}^{4} m_i$ must be

Answer Key

Q1 (3) **Q2** (4) **Q3** (2) **Q4** (1)

Q5 (1) **Q6** (2) **Q7** (3) **Q8** (1)

Q9 (4) **Q10** (1) **Q11** (3) **Q12** (4)

Q13 (1) **Q14** (1,2) **Q15** (2,4) **Q16** (1,2,4)

Q17 (1,3) **Q18** (2,3) **Q19** (1,3,4) **Q20** (2,3,4)

Q21 (1,2,4) **Q22** (1,3) **Q23** (2,3) **Q24** (1,2,3,4)

Q25 (3,4) **Q26** (2) **Q27** (3) **Q28** (3)

Q29 (4) **Q30** (3) **Q31** (3) **Q32** (2)

Q33 (3) **Q34** (4) **Q35** (4) **Q36** (4)

Q37 (1) **Q38** (2) **Q39** (5) **Q40** (5)

Q41 (1) **Q42** (0) **Q43** (5) **Q44** (4)

Q45 (5) **Q46** (7) **Q47** (2) **Q48** (3)

Q49 (9) **Q50** (1)